

Erdős Problem #866

EP-866

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 3$ and $g_k(N)$ be minimal such that if $A \subseteq \{1, \dots, 2N\}$ has $|A| \geq N + g_k(N)$ then there exist integers $b_1, \dots, b_k$ such that all $\binom{k}{2}$ pairwise sums are in A (but the b_i themselves need not be in A). Estimate $g_k(N)$.